

Paper VII — Gravity as a Lattice Dielectric: An Impedance-Matched Renormalisation of the Field Rotation Rate, GR-Identical PPN, and a Structural Newton Constant

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Series: “A Universe in a Bottle” — Paper VII of XI. Builds on Paper I (light as rotation rate), Paper II (the cell-size co-certification), and completes the four-interaction set (IV–VII). Its strong-field limit is the exact Schwarzschild/Kerr solution of the induced Einstein equation (F178); the earlier horizon-free “dielectric black hole” treatment (deprecated, F114) is reclassified as the PPN-order representation (see §4.2).

*Revision 2 (2026-06-08): the SI cell is updated throughout to the **adopted canonical value** of Finding F107, $a = \sqrt{8\pi} 3^{1/4} \ell_P = 6.5978 \ell_P = 1.0664 \times 10^{-34} \text{ m}$ (tick $\tau = 2.0537 \times 10^{-43} \text{ s}$). Revision 1 quoted an inconsistent “ $\approx 3.8 \ell_P$, one generation” in one place; that is corrected here — the mode count is $g_* = 48$ (three generations $\times$ 16 Weyl fields), giving $a/\ell_P = \sqrt{8\pi} 3^{1/4} = 6.5978$.*

*Revision 3 (2026-06-30): **strong-field section re-issued per Finding F178**. The model now adopts the **induced Einstein equation** $G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$ as the canonical gravitational law, sourced by the **full** stress-energy tensor. The single impedance-matched dielectric $K = e^{2u}$ is reclassified as the **vacuum/weak-field (PPN-order) representation** of that law — exact in spirit for lensing, PPN $\beta = \gamma = 1$, redshift and the Newton-constant origin story, but **not** the exact strong-field metric. The canonical strong-field object is therefore **exact Schwarzschild (and Kerr for rotation)**, with a genuine horizon; the horizon-free “dielectric black hole” (deprecated) is demoted to a representation artifact (F114→F178). Inside matter the metric carries two independent functions and the dynamics are GR/TOV (the covariant interior kernel, F181). The weak-field claims of §§2–3, 5–6 are unchanged; §4 is rewritten below.*

Abstract

We present gravity in the BCC quantum-cellular-automaton model as a single, impedance-matched **lattice dielectric** $K(\mathbf{x})$: a position-dependent renormalisation of the $(\mathbf{E}, \mathbf{B})$ rotation rule that Paper I identified with the speed of light. With the canonical index $K = \exp(2GM/rc^2)$, metric legs $A = 1/K$, $B = K$ obeying the exact reciprocal lock $AB \equiv 1$, the model reproduces general relativity at the post-Newtonian level: $\beta = \gamma = 1$, Mercury perihelion advance $42.98''/\text{century}$, and full Einstein factor-2 light bending $(4GM/bc^2)$. A single scalar achieves what a single scalar generically cannot — simultaneous factor-1 redshift and factor-2 deflection — because the reciprocal lock keeps the lattice impedance $\sqrt{\mu/\varepsilon} = 1$ exactly, so the rotation stays a *proper* rotation with no scalar contamination. We derive the dielectric placement $\varepsilon = \mu = K$ from the proper-rotation requirement plus the conformal invariance of source-free Maxwell, show that light dynamically bends light (the rest-mass route gives zero), promote the potential to a causal dynamical field, and — crucially — **free Newton’s constant from the Sakharov premise**: the gravity field carries zero tree stiffness because the dominant lattice stress tensor is traceless, so the induced (loop) channel is forced as a theorem, giving the closed form $G = a^2 c^3 / (8\pi \sqrt{3} \hbar)$ and the parameter-free prediction $a/\ell_P = \sqrt{8\pi} 3^{1/4} = 6.598$. Sixteen field-level and dynamical tests pass; the SI cell is gated by two independent campaigns (absolute lensing and the GRB time-of-flight bound). **The dielectric is the vacuum/weak-field representation of the model’s canonical gravitational law — the**

induced Einstein equation $G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$ **sourced by the *full* stress-energy tensor (§4, Revision 3 / F178)**: a T^{00} -only source is not Lorentz covariant, and an impedance-locked single scalar cannot source an isotropic perfect fluid (it forces $p_r = -p_t$, F173). The exact strong-field object is therefore **Schwarzschild/Kerr** (with a horizon) and the interior is GR/TOV (F181), so the horizon-free “dielectric black hole” is reclassified as a PPN-order artifact. The model’s lasting gravitational content is the **origin and structural value of G** , with low-energy dynamics that are exactly Einstein’s.

1. Introduction

Gravity is the interaction least like the others: it couples universally, it is geometric in general relativity (GR), and its quantisation is unsolved. On a fixed lattice one cannot literally curve space, so a faithful model must encode the metric some other way. Paper I provides the handle: the speed of light is the rotation rate of the real field pair, $c_{\text{lat}} = d\Omega/d|\mathbf{k}|$. If that rate is made *position-dependent*, light slows in a well and bends — gravity becomes a renormalisation of the rotation rule itself.

This is the polarizable-vacuum / dielectric line of Puthoff and of Ostoma–Trushyk, and it is native to the present model because (Papers II–III) mass itself is confined (**E, B**) rotation — there is no separate “mass substance” to source a metric from, only confined electromagnetic energy. The result is **one field, one source** (total field energy), in contrast to the two-leg, rest-mass-sourced metric that the earlier emergent-gravity forks of this project required for factor-2 bending (Findings F50/F52/F62, now superseded; their lessons are retained).

2. The dielectric and its observables

We write the static isotropic metric as $ds^2 = -A c_0^2 dt^2 + B \delta_{ij} dx^i dx^j$, with photon coordinate speed $c_{\text{eff}} = c_0 \sqrt{A/B}$ and eikonal index $n = c_0/c_{\text{eff}} = \sqrt{B/A}$. With $u \equiv GM/(rc^2) = -\phi/c^2$, the two solar-system observables are the leading u -slopes

$$Z \equiv -\left. \frac{d\sqrt{A}}{du} \right|_0 \quad (\text{GR: } 1), \quad K_{\text{bend}} \equiv 2 \left. \frac{dn}{du} \right|_0 \quad (\text{GR: } 4, \text{ Newton: } 2). \quad (2.1)$$

2.1 One scalar gives both — the reciprocal lock

The decisive go/no-go: can one scalar give factor-1 redshift *and* factor-2 bend simultaneously? Three single-scalar placements (exact series):

placement	A, B	Z	K_{bend}	impedance const?	passes GR?
dielectric	$A = 1/K, B = K$	1	4	yes ($= 1$)	yes
clock-only (rest leg)	$A = (1 - u)^2, B = 1$	1	2	no	no
refractive-only (Gordon)	$A = 1, B = (1 + u)^2$	0	2	no	no

Only the dielectric passes both. The structural reason is the **reciprocal lock** $AB = 1$: a genuine dielectric scales $\varepsilon = \mu = K$ together, so the impedance $\sqrt{\mu/\varepsilon} = 1$ is exactly u -independent —

the $(\mathbf{E}, \mathbf{B})$ amplitude ratio is preserved and the rotation stays a *proper* rotation with no scalar contamination. Fixing K by factor-1 redshift ($\sqrt{A} = K^{-1/2} = 1 - u$) then forces $n = \sqrt{B/A} = K = 1 + 2u + O(u^2)$, hence $K_{\text{bend}} = 4$ — Einstein bending from one field.

2.2 Radiation gravitates as rest mass

The empirically sharp discriminator: two equal-energy sources — a rest-mass blob and a massless spherical shell of standing $(\mathbf{E}, \mathbf{B})$ field energy — bend probe rays equally under the dielectric coupling (ratio 1.00002), because both carry the same monopole. Under the rest-mass coupling the massless shell sources **nothing**. This zero-versus-Einstein split is the model’s sharpest gravitational prediction, and it is realised dynamically: a 2D Maxwell pulse propagating through a dielectric well sourced by a *massless* field-energy lump bends with full factor-2, where the rest-mass route gives zero (light bends light).

3. Deriving the dielectric placement

The one posited input — *that* the renormalisation enters as a genuine dielectric rather than as a clock-only or refractive-only scalar — is derived in four exact steps from the proper $(\mathbf{E}, \mathbf{B})$ -rotation law (Paper I) plus a lattice confirmation:

1. **Plébański equivalent medium.** The equivalent medium of a static isotropic metric is $\varepsilon = \mu = \sqrt{B/A} = K$.
2. **Conformal invariance.** Source-free Maxwell is conformally invariant in $3 + 1\text{D}$, so the EM sector sees only the conformal class; it fixes $\varepsilon = \mu = K$ but is blind to the conformal factor.
3. **Impedance + index.** Demanding a proper rotation ($Z = \sqrt{\mu/\varepsilon} = 1$, no scalar component) *and* index $n = \sqrt{\varepsilon\mu} = K$ has the unique solution $\varepsilon = \mu = K$.
4. **Reciprocal lock.** The conformal factor — the one thing EM is blind to — is supplied by the measured factor-1 redshift $\sqrt{A} = 1 - u$, giving $A = (1 - u)^2$, $B = K$, $AB = 1$.

A 1D Yee finite-difference simulation confirms it: the impedance-matched dielectric is reflectionless at an index step (to the grid floor, $R = 0.009$), while refractive-only and clock-only placements reflect at the Fresnel value ($R = 0.111$). The reflected backward wave *is* the scalar contamination a proper rotation forbids — so “position-dependent rotation that stays a proper rotation” forces the dielectric (Finding F64, derivations D-EM1–D-EM5).

4. Strong field: the canonical law is the induced Einstein equation (re-issued, F178)

4.1 PPN — where the dielectric is exact

Exact post-Newtonian analysis of $g_{tt} = -A$, $g_{ij} = B\delta_{ij}$:

metric	β	γ	light bend/GR	Mercury /cy
redshift-fixed $K = (1 - u)^{-2}$	$\frac{1}{2}$	1	1	50.1 (excluded)
exponential $K = e^{2u}$	1	1	1	42.98
Schwarzschild (ref)	1	1	1	42.98

Light deflection depends only on $\gamma = 1$, so the dielectric passes the bending test exactly. Perihelion advance depends on β : the naive linear dielectric $K = (1 - u)^{-2}$ has $\beta = \frac{1}{2}$ (Mercury 50.1"/cy, a 16.7% excess excluded by MESSENGER), while the impedance-matched exponential

$$K = e^{2GM/rc^2}, \quad A = 1/K, \quad B = K, \quad AB \equiv 1, \quad (4.1)$$

restores $\beta = \gamma = 1$ (derivation D-EM9). To **PPN order** the dielectric is GR-identical, and *this is the regime in which it is canonical*: all weak-field/solar-system observables — light bending, Mercury, Shapiro delay, gravitational redshift — are reproduced exactly, with the reciprocal lock $AB \equiv 1$ doing the work of two independently-sourced legs.

4.2 The canonical strong-field law (F178)

The exponential (4.1) is, however, only the **vacuum/weak-field representation** of the model's gravity, not its exact strong-field metric. Finding F178 makes the canonical law the **induced Einstein equation**

$$\boxed{G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}, \quad G = \frac{a^2 c^3}{8\pi\sqrt{3}\hbar}} \quad (4.2)$$

sourced by the **full** stress-energy tensor. Three independent reasons force this over the single energy-sourced scalar:

1. **Lorentz covariance.** A source built from the energy density T^{00} alone is not a tensor equation; only coupling to the full $T_{\mu\nu}$ is frame-independent. The single scalar can only ever be a static, preferred-frame approximation.
2. **The single scalar carries anisotropic stress.** The exact Einstein tensor of (4.1) has $p_r = -p_t = -e^{-2u}u'^2/8\pi$ (F173, sympy-exact): an impedance-locked single scalar **cannot** source an isotropic perfect fluid, so it cannot be the field equation inside matter.
3. **Neutron stars.** The literal energy-only law has no maximum mass; the full-tensor source recovers TOV structure and the observed $\sim 2 M_\odot$ ceiling (F176/F181).

The consequences of (4.2):

- **Vacuum / exterior.** The exact spherically-symmetric vacuum solution is **Schwarzschild** (Kerr for rotation), *with a horizon*. The exponential $K = e^{2u}$ agrees only to PPN order: the two metrics differ at $O(u^2)$ (e.g. the spatial leg, $e^{2u} = 1 + 2u + 2u^2 + \dots$ vs isotropic Schwarzschild $(1 + u/2)^4 = 1 + 2u + \frac{3}{2}u^2 + \dots$), and the dielectric lapse $A = e^{-2u} > 0$ never vanishes — it is **horizon-free**. That horizon-free “dielectric black hole” (the subject of the earlier, now-deprecated F114 treatment) is therefore reclassified as an artifact of treating the PPN-order representation as fundamental; the canonical black hole is Schwarzschild/Kerr.
- **Interior.** Inside matter the metric carries its **second independent function** (A, B free, $AB \neq 1$) and the dynamics are GR/TOV. The covariant two-function interior kernel (F181) solves the full isotropic perfect-fluid Einstein equations exactly where the single scalar cannot, reproducing GR-TOV to the integrator floor.

4.3 What is preserved

The reclassification is a deliberate trade — Lorentz covariance and automatic consistency with neutron stars, gravitational waves and cosmology, in exchange for the dielectric’s distinctive strong-field phenomenology. What survives intact is everything in the regime where the dielectric *is* the representation of (4.2): factor-2 light bending, $\beta = \gamma = 1$, Mercury, Shapiro, redshift; the structural Newton constant $G = a^2 c^3 / (8\pi\sqrt{3}\hbar)$ (§5); and the emergent-origin story (gravity as an induced renormalisation of the $(\mathbf{E}, \mathbf{B})$ rotation rate). A causal dynamical completion ($\square\Phi = -4\pi G\rho$ in the weak field; the full spin-2 equation in general) gives retarded propagation at the graviton speed $c_g = c_{\text{lat}} = 1/\sqrt{3}$ (F180). The dynamical Dirac wave-packet battery — equivalence principle, redshift $\propto \sqrt{-g_{tt}}$, factor-2 deflection, unitary backreaction — reproduces every result on the **two-function** background without modification (F181), so nothing in the weak-field phenomenology is lost by abandoning the impedance lock.

5. Newton’s constant from lattice structure

The last posited input, the $4\pi G$ coupling, is fixed by recognising what the gravity field *is* (Finding F79).

5.1 The pivot: K is a conformal factor, not a fundamental field

By the derivation of §3, K is the conformal factor the EM action is blind to. A conformal (Weyl) rescaling couples to matter through the *trace* of the stress tensor, $\delta S = \int \sqrt{g} T^\mu{}_\mu \delta\sigma$. The source-free electromagnetic stress tensor is **traceless in $3 + 1\mathbf{D}$** , $T^\mu{}_\mu^{(\text{EM})} = 0$ (verified to 3.6×10^{-15} on random fields). Therefore the conformal factor has **no source and no tree action** — the gravity field has *zero tree stiffness*. There is no fundamental graviton on this lattice whose bare kinetic term could set G ; K is a derived reparametrisation, and its entire stiffness is the induced (loop) response of the fundamental $(\mathbf{E}, \mathbf{B})$ /spinor modes.

5.2 The loop channel is forced

The two candidate stiffnesses scale differently in c_{lat} : the tree (fundamental-graviton) channel as c_{lat}^{+2} , the induced (loop) channel as c_{lat}^{-1} , a gap of c_{lat}^3 . Since §5.1 shows there is no tree term to measure, the physical coupling is the loop channel,

$$\frac{1}{16\pi G} \propto \frac{1}{c_{\text{lat}}} = \sqrt{d}. \quad (5.1)$$

5.3 Every input is structural; the closed form

input	value	fixed by
c_{lat}	$1/\sqrt{d}$	BCC rotation rule (Paper I)
η_{Weyl}	$\frac{1}{12}$	Seeley–DeWitt heat-kernel coefficient

input	value	fixed by
g_*	$48 = 16 \times 3$	O_h generation theorem (Paper VIII) $\times$ anomaly-free content

The mode count g_* is the decisive upgrade: the generation count 3 is a theorem about the BCC point group ($\dim T_{1u}$, no 4-dim single-valued irrep), so with the anomaly-free 16-Weyl content per generation ($L=2, e_R=1, Q=6, u_R=3, d_R=3, \nu_R=1$), $g_* = 48$ is a count of the lattice’s own protected normal modes, not a matter-content input. Assembling the Sakharov integral gives the closed form

$$\boxed{\frac{1}{G} = 2\pi \eta g_* \sqrt{d} \frac{\hbar}{a^2 c^3}, \quad G = \frac{a^2 c^3}{8\pi \sqrt{3} \hbar}}, \quad (5.2)$$

with the parameter-free coefficient $2\pi \eta g_* \sqrt{d} = 2\pi \cdot \frac{1}{12} \cdot 48 \cdot \sqrt{3} = 8\pi \sqrt{3} = 43.531$. The dimensionless content the lattice predicts is

$$\frac{a}{\ell_P} = \sqrt{2\pi \eta g_*} d^{1/4} = \sqrt{8\pi} 3^{1/4} = 6.5978, \quad (5.3)$$

with tick $\tau/t_P = \sqrt{8\pi} 3^{-1/4} = 3.809$ and the d -independent invariant $\sqrt{ac\tau}/\ell_P = \sqrt{8\pi} = 5.013$. In SI, $a = 6.5978 \ell_P = 1.0664 \times 10^{-34}$ m and $\tau = 2.0537 \times 10^{-43}$ s (Option C lightcone $a/\tau = c\sqrt{3}$). Anchoring a at ℓ_P returns $G = 6.6743 \times 10^{-11}$ SI to 3×10^{-8} (CODATA round-off), confirming the algebra.

5.4 The one honest limit

No theory yields a dimensionful G from pure numbers — that needs one ruler. The lattice predicts the *dimensionless* a/ℓ_P (equivalently $8\pi\sqrt{3}$). To state G in SI one must anchor the single length a . The canonical adoption (Finding F107) takes the F79 value $a = \sqrt{8\pi} 3^{1/4} \ell_P$ as the ruler and demotes the F83 fermion-mass map from anchor to consistency check (the canonical a sits ~ 16.5 decades below the top-quark mass ceiling). The genuinely independent alternative anchor remains a measured fermion mass via the lattice-mass map (Paper III), which references no Planck length; under it G becomes an output of particle data. The spin-1 gauge contribution to g_* is a bounded, mild $\sqrt{\cdot}$ correction.

5.5 Why the canonical cell was adopted (the gating)

Adoption was gated on two passed campaigns (Finding F107, 6/6):

1. **L4 absolute lensing.** On the canonical $K = e^{2u}$ the straight-ray bending coefficient is $K_{\text{bend}} = -4$ exact at all field strengths; the lattice absolute coefficient is 4; the G -match and the solar-limb deflection $1.7512''$ both hold at 3.0×10^{-8} .
2. **The GRB gate.** The even-channel $n = 2$ Lorentz-violation scale is $E_{\text{QG},2} = \sqrt{54} \hbar c/a$. At the F79 value it clears LHAASO by 1.9×10^7 ; at the F83 mass-ceiling cell it would be excluded by ~ 9 decades. The bound thus observationally *selects* the F79 cell. Falsification handle: any $n = 2$ time-of-flight bound above 1.36×10^{19} GeV kills the adopted cell.

Every SI prediction in the series (the canonical Schwarzschild/Kerr shadow per §4.2, Paper X nuclear scales once calibrated) now evaluates at this single a (Finding F112).

6. Cell-size co-certification with the photon

The cell size that (5.3) forces, $a = 6.5978 \ell_P$, is the same cell size at which the *chiral* photon was excluded by polarimetry. The **paired photon** (Paper II) removes that birefringence prediction entirely, so the gravity-fixed cell size now simultaneously (i) reproduces G and (ii) survives all astrophysical vacuum-birefringence bounds, with the Lorentz-violation cutoff at $E_{\text{QG},2} \approx 1.36 \times 10^{19}$ GeV. Papers II and VII thus co-certify a single cell size — a consistency gain that does not touch any dimensionless prediction.

7. Verification summary

Result	Section	Residual / status
Single-scalar viability ($Z = 1$, $K_{\text{bend}} = 4$)	§2.1	exact; ratio to rest-leg = 2.000
Radiation gravitates as rest mass	§2.2	1.00002 (rest-leg: 0)
Dielectric placement derived (reflectionless)	§3	$R = 0.009$ (Fresnel 0.111 otherwise)
Absolute 3D bend $K_{\text{bend}} \rightarrow 4$	§4	4.0015 at $u = 10^{-4}$
PPN $\beta = \gamma = 1$ (Mercury 42.98")	§4	exact (vs 50.1" for linear)
EM stress tensor traceless	§5.1	3.6×10^{-15}
Channel exponents $(2, -1, -3)$	§5.2	exact
$g_* = 48$; $a/\ell_P = \sqrt{8\pi}3^{1/4} = 6.5978$	§5.3	exact; G consistent to 3×10^{-8}
L4 lensing + GRB gate (adoption)	§5.5	6/6; 3.0×10^{-8}

Sixteen field-level and dynamical tests pass. Underlying findings: F64 (EM-connection / dielectric gravity, D-EM1–D-EM11), F79 (structural Newton constant), F107 (canonical a adoption / GRB gate), F112 (SI predictions), F55–F62 (the superseded two-leg route and its lessons), F69 (cell-size co-certification). Strong-field re-issue (§4.2): F178 (full-tensor source adopted), F173 (single-scalar anisotropy), F176/F181 (covariant interior / TOV), F180 (graviton speed), F114 (the demoted dielectric black hole).

8. Discussion

Gravity in this model is not a new field but a modulation of the field already present: the rotation rate that *is* the speed of light. In the **weak field** one impedance-matched scalar reproduces GR's

PPN phenomenology, with the reciprocal lock $AB = 1$ doing the work that two independently-sourced metric legs did in the earlier route. The canonical law, however, is the **induced Einstein equation** $G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$ sourced by the full stress-energy tensor (§4.2, F178): the dielectric is its vacuum/weak-field representation, the exact strong-field object is Schwarzschild/Kerr, and the interior is GR/TOV (F181). The deepest result is that Newton’s constant is not free: because the gravity field is a conformal reparametrisation with no tree stiffness, G is the lattice’s induced dielectric stiffness against bending its own rotation rule, fixed (up to one ruler) by $c_{\text{lat}} = 1/\sqrt{3}$, $\eta = \frac{1}{12}$, and the generation theorem $g_* = 48$. The model’s distinctive gravitational content is thus the **origin and value of G** (and the rotation-rate origin story), with low-energy dynamics that are exactly Einstein’s.

Open items, all bounded: the spin-1 gauge contribution to g_* ; the Kerr/frame-dragging sector on the two-function kernel; a two-body / gravitational-wave sector (graviton speed $c_g = c_{\text{lat}}$ confirmed, F180); and 3D dynamical light-bends-light. The lineage is Sachs’ electrogravity (the notebook’s starting point), Puthoff’s polarizable vacuum, Yilmaz’s exponential metric, and Ostoma–Trushyk — now understood as the **weak-field representation** of induced (Sakharov-type) Einstein gravity.

References

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5. M. Sachs, *General Relativity and Matter* (Reidel, 1982) — electrogravity unification (notebook source).
6. C. M. Will, *Theory and Experiment in Gravitational Physics* (Cambridge, 2018) — PPN formalism.
7. Project findings: F64 (dielectric gravity), F79 (structural G), F107 (canonical cell / GRB gate), F112 (SI predictions), F55–F62 (two-leg route), F69 (cell-size co-certification).

Companion papers: I (light as rotation), II (photon / cell size), VIII (generation count / g_).*